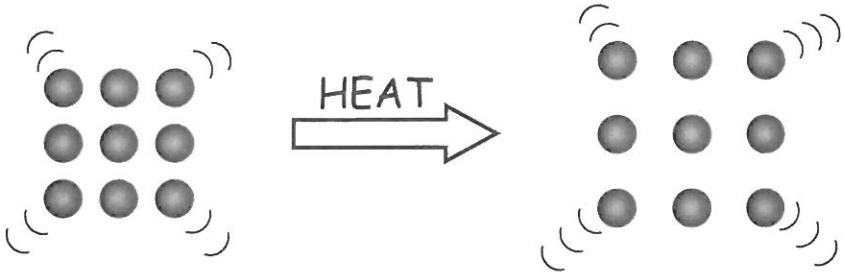
W.S.47. **Expansion.** Name .....

If a metal bar is heated up it EXPANDS (gets bigger) slightly. This happens because the metal particles gain more energy and vibrate more.

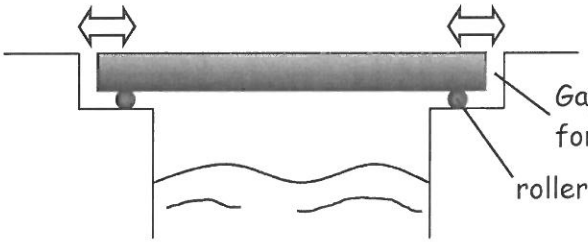


The particles vibrate more and so they move further apart.

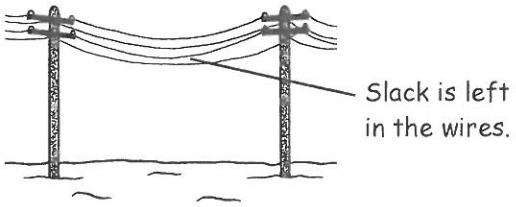
Most materials expand slightly when they are heated.

Problems caused by expansion.

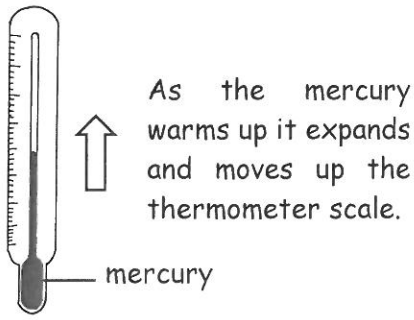
In hot weather a bridge could expand and buckle.



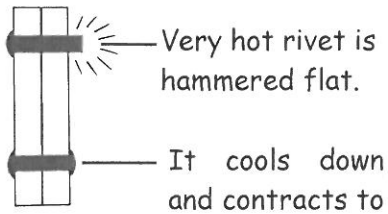
Overhead wires shorten in cold weather and could snap.



Uses of expansion.



Rivets hold metal plates tightly together.



Exercise - Fill in the missing words in the passage below.

Most materials ..... when they are heated and ..... when they are cooled. This is because their particles ..... more when hot and so move further ..... In hot weather a metal bridge could expand and ..... To stop this from happening it is held on rollers. Overhead wires could contract and ..... in cold weather. To stop this from happening they are given slack when they are put up. Mercury is a liquid metal that is used inside a ..... When it is put in a warmer place the mercury expands and moves up the .....

vibrate **apart** **snap** **thermometer** **contract** **buckle** **expand** **scale**

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